Testy hypotéz pre parametre normálneho rozdelenia

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```
library(latex2exp)# kniznica na vkladanie TeX textu do grafov
library(latexpdf)
```

Testy hypotez pre parametre normalneho rozdelenia

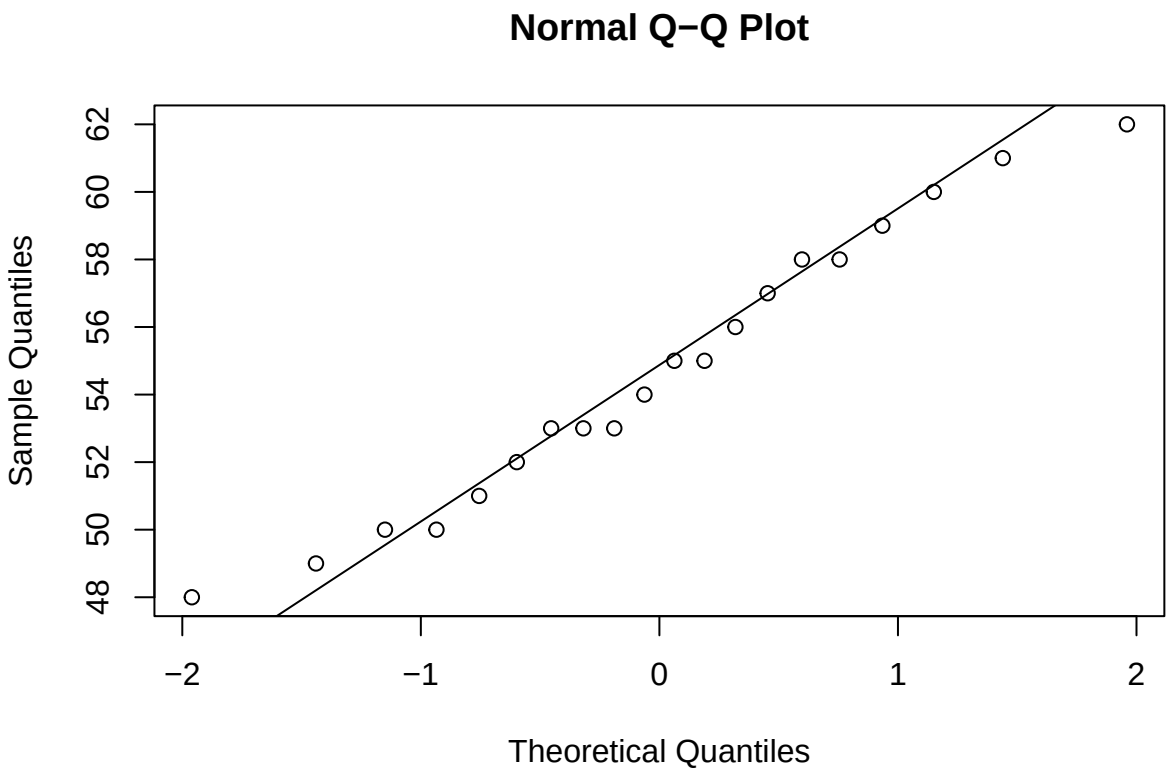
Test pre strednu hodnotu μ a σ pozname.

Priklad Udaje o dlzke casu v sekundach, ktory stravia pouzivatelia na www stranke, disperzia je znama $\sigma = 3.8$. Testujte hypotezu, ze klient stravi na stranke v priemere 55s. Hladina vyznamnosti $\alpha = 0.05$ Tvar hypotez, overenie normality

$$H_0 \quad \mu = 55 \qquad H_1 \quad \mu \neq 55$$

Grafy na overenie zhody rozdelenia (default normalne rozdelenie)

```
cas <- c(48, 55, 51, 62, 53, 58, 60, 50, 49, 57, 52, 61, 54, 56, 59,53,
         50, 58, 55, 53)
qqnorm(cas)# kvantil kvantil graf
qqline(cas)# pridat ciaru
```



Shapiro Wilk test

```
shapiro.test(cas)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  cas
## W = 0.96958, p-value = 0.7459
```

P hodnota=0.74>0.05, nezamietame hypotezu o normalite dat. Pozname disperziu, teda Z test kniznica DescTools

```
library(DescTools)
ZTest(cas,mu=55,sd_pop=3.8)
```

```
##
##  One Sample z-test
##
## data:  cas
## z = -0.35306, Std. Dev. Population = 3.8, p-value = 0.724
## alternative hypothesis: true mean is not equal to 55
## 95 percent confidence interval:
##  53.03461 56.36539
## sample estimates:
## mean of x
##      54.7
```

```
ZTest(cas,mu=55,sd_pop=3.8)$p.value
```

```
## [1] 0.7240409
```

Rozhodnutie podla PValue=0.72>0.05, nezamietam H_0. Ano, klient stravi na stranke v priemere 55s Moznosti: Kniznica broom, ktora umozni dat prehladnejsi vystup

```
library(broom)
broom::tidy(ZTest(cas,mu=55,sd_pop=3.8))
```

```
## # A tibble: 1 x 8
##   estimate statistic p.value parameter conf.low conf.high method     alternative
##      <dbl>     <dbl>   <dbl>     <dbl>    <dbl>     <dbl> <chr>      <chr>
## 1     54.7    -0.353   0.724       3.8     53.0      56.4 One Sampl~ two.sided
```

Nova hypoteza: Otazka v skutocnosti znela, ci zotrvaju na stranke aspon 58s (dokazne bremeno je na alternative).

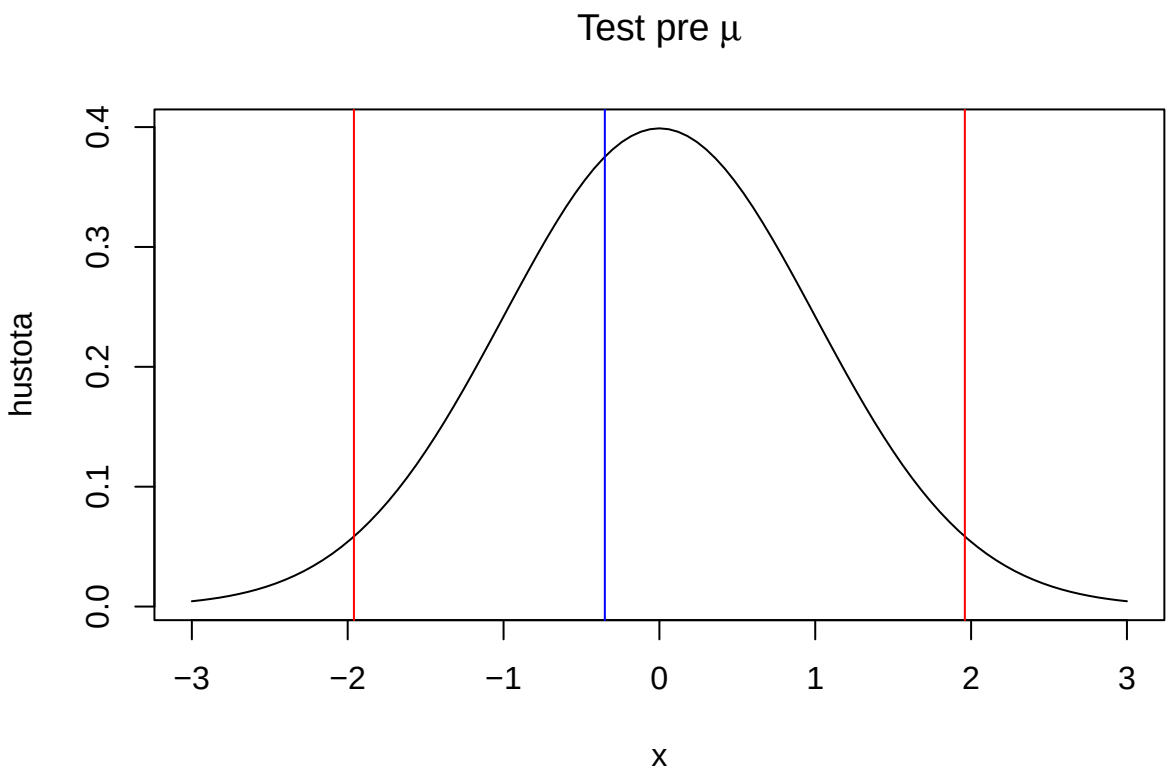
$$H_0 \quad \mu = 58 \qquad \mu > 58$$

```
ZTest(cas,mu=58,alternative = "g",sd_pop = 3.8)
```

```
##
##  One Sample z-test
##
## data:  cas
## z = -3.8837, Std. Dev. Population = 3.8, p-value = 0.9999
## alternative hypothesis: true mean is greater than 58
## 95 percent confidence interval:
##  53.30236      Inf
## sample estimates:
## mean of x
##      54.7
```

Phodnota=0.99>0.05, nezamietame H_0, nie je pravda, ze zotrvaju viac ako 58 sekund. Ak by sme testovali napriklad na hladine vyznamnosti $\alpha = 0.1$, aj tak rozhodujeme : ak Phodnota>0.1, tak nezamietame nulovu hypotezu. Test graficky, ten prvy test o rovnosti.

```
plot(function(x)dnorm(x,0,1),from=-3, to=3,
     main=TeX("Test pre $\\mu$"),
     ylab="hustota")
q <- qnorm(0.975,0,1)# kvantil normalneho rozdelenia, hranica
z <- -0.35# statistika z vystupu testu
abline(v=z,col="blue")
abline(v=q,col="red")
abline(v=-q,col="red")
```



Test pre μ ak σ nepozname

Testujeme testy z predosleho prikladu za predpokladu, σ nepozname

```
t.test(cas,mu=55)
```

```
##
##  One Sample t-test
##
## data:  cas
## t = -0.32682, df = 19, p-value = 0.7474
## alternative hypothesis: true mean is not equal to 55
## 95 percent confidence interval:
##  52.77871 56.62129
## sample estimates:
## mean of x
##      54.7
```

```
t.test(cas,mu=55)$p.value
```

```
## [1] 0.7473805
```

Phodnota>0.05 nezamietame H_0 tvrdenie plati. Druha hypoteza

```
t.test(cas,mu=58,alternative = "g")
```

```
##
##  One Sample t-test
```

```
##
## data:  cas
## t = -3.595, df = 19, p-value = 0.999
## alternative hypothesis: true mean is greater than 58
## 95 percent confidence interval:
##  53.11274      Inf
## sample estimates:
## mean of x
##      54.7
```

Phodnota>0.05, nezamietame H_0, teda nie je pravda, ze vydrzia viac ako 58 sekund.

Plnicka jablkovej stavy

Firma predava jablkovu stavu v 0.5l baleniach. Plniaca linka presla servisom. Potom sme namerali tieto hodnoty plneneho objemu v ml. Na hladine vyznamnosti $\alpha = 0.1$ testujte hypotezu, ze linka je dobre nastavena.

$$H_0 \quad \mu = 500 \qquad H_1 \quad \mu \neq 500$$

```
jablko <- c(499.2,496.8,502.1,498.5,501,503,500.7,
            501.5,501.8,499.1,500.9,502.2,501.7, 500.4,
            500.2,501.1,499.9,500.2,501.1,500.8,499.3)
shapiro.test(jablko)# normalne rozdelene, ok
```

```
##
##  Shapiro-Wilk normality test
##
## data:  jablko
## W = 0.96169, p-value = 0.5507
```

```
t.test(jablko,mu=500)# nemusim dat hladinu vyznamnosti
```

```
##
##  One Sample t-test
##
## data:  jablko
## t = 1.7663, df = 20, p-value = 0.09261
## alternative hypothesis: true mean is not equal to 500
## 95 percent confidence interval:
##  499.9009 501.1944
## sample estimates:
## mean of x
##  500.5476
```

Kedze $\alpha = 0.1$ a Phodnota=0.09<0.1 zamietame H0. Ak by bola 0.05, .09>0.05 tak by sme nezamietli. Zvysili sme chybu prveho druhu na 10%.

Zlozene alternativy

Firma, ktora vyraba baterie do notebookov tvrdi, ze jednu vyrobi do 13 minut. Overte toto tvrdenie, ak mate k dispozicii 20 merani, $\alpha = 0.05$. Stanovte spravne nulovu a alternativnu hypotezu.

```
baterie <- c(12,19,16,13,15,12,14,20,15,19,17,20,
             13,9,11,20,12,19,8,13)
shapiro.test(baterie)#normalne rozdelene
```

```
##
##  Shapiro-Wilk normality test
##
## data:  baterie
## W = 0.93108, p-value = 0.1619
```

$$H_0 \quad \mu = 13 \qquad H_1 \quad \mu < 13$$

```
t.test(baterie,mu=13,alternative = "l")
```

```
##
##  One Sample t-test
##
## data:  baterie
## t = 2.2006, df = 19, p-value = 0.9798
## alternative hypothesis: true mean is less than 13
## 95 percent confidence interval:
##      -Inf 16.30361
## sample estimates:
## mean of x
##     14.85
```

Phodnota=0.97>0.05, nezamietame nulovu hypotezu, neprijimame alternativy, trva mu to viac, odhad strednej hodnoty je 14.85.

Vyrobca tvrdi, ze baterky do elektornickych pristrojov vydrzia aspon 19 hodin nepretrzitej prevadzky. Testujte, ak mame k dispozicii datovy subor baterie1. Ak α dalej neuvedieme, bude 0.05.

```
baterie1<- c(18.2,19.6,18.6,19.4,17,18.5,18,18.4,19,
             18,17.9,18.1)
shapiro.test(baterie1)#normalne rozdelene
```

```
##
##  Shapiro-Wilk normality test
##
## data:  baterie1
## W = 0.95612, p-value = 0.7274
```

$$H_0 \quad \mu = 19 \qquad H_1 \quad \mu > 19$$

```
t.test(baterie1,mu=19,alternative = "g")
```

```
##
##  One Sample t-test
##
## data:  baterie1
## t = -2.9777, df = 11, p-value = 0.9937
## alternative hypothesis: true mean is greater than 19
## 95 percent confidence interval:
##  18.02478      Inf
## sample estimates:
```

```
## mean of x
##  18.39167
```

Phodnota>0.05, nezamietame nulovu hypotezu nevydrzia viac ako 19 hodin.

```
t.test(baterie1,mu=19,alternative = "l")#len ci dojde k sporu
```

```
##
##  One Sample t-test
##
## data:  baterie1
## t = -2.9777, df = 11, p-value = 0.006285
## alternative hypothesis: true mean is less than 19
## 95 percent confidence interval:
##      -Inf 18.75856
## sample estimates:
## mean of x
##  18.39167
```

Ak si obratime hypotezy, vacsinou nie je spor. Tu H0 zamietame prijame H1, co nie je spor vydrzia menej ako 19.

Priklad

Pizzeria ABC ma na letaku uvedene, ze pizzu dovezu do 30 min. Casy dodavok su v datovom subore pizza. Testujte na hladine vyznamnosti $\alpha = 0.05$. Vyslovte zaver.

```
pizza <- c(27,25,30,28,29,24,30,26,28,32,
           24,32,31,29,28,29,35,34,30,31,18,22)
```

$$H_0 \quad \mu = 30 (\geq 30) \quad H_1 \quad \mu < 30$$

```
t.test(pizza,mu=30,alternative="l")
```

```
##
##  One Sample t-test
##
## data:  pizza
## t = -2.0426, df = 21, p-value = 0.02692
## alternative hypothesis: true mean is less than 30
## 95 percent confidence interval:
##      -Inf 29.72782
## sample estimates:
## mean of x
##  28.27273
```

Testy pre disperziu normalneho rozdelenia

V prvom priklade sme tvrdili, ze $\sigma = 3.8$, pre datovy subor cas. Na hladine vyznamnsoti $\alpha = 0.05$ overte toto tvrdenie.

$$H_0 \quad \sigma^2 = 3.8^2 \qquad H_1 \quad \sigma^2 \neq 3.8^2$$

Knznica DescTool, EnvStats

```
EnvStats::varTest(cas,sigma.squared = 3.8^2)
```

```
## $statistic
## Chi-Squared
##    22.17452
##
## $parameters
## df
## 19
##
## $p.value
## [1] 0.5513144
##
## $estimate
## variance
## 16.85263
##
## $null.value
## variance
##    14.44
##
## $alternative
## [1] "two.sided"
##
## $method
## [1] "Chi-Squared Test on Variance"
##
## $data.name
## [1] "cas"
##
## $conf.int
##       LCL       UCL
##  9.746646 35.951205
## attr(,"conf.level")
## [1] 0.95
##
## attr(,"class")
## [1] "htestEnvStats"
```

```
EnvStats::varTest(cas,sigma.squared = 3.8^2)$p.value
```

```
## [1] 0.5513144
```

Nezamietame H_0. Tvrdenie je pravdive.

Dvojvyberove testy

Merania idu v pare, vytvorime diferencie a tie zmysluplne testujeme. Su dane casy v sekundach, pred a po, pocas ktorych riesili ziaci kontrolne ulohy pred a po cviceniach z pamatoveho pocitania. Zlepsili cvicenia schopnost ziakov rychlejsie riesit ulohy?

$$H_0 \quad \mu_{pred} - \mu_{po} = 0 \qquad H_1 \mu_{pred} - \mu_{po} > 0$$

```r
pred <- c(87,61,98,90,93,74,83,72,81,75,83)
po <- c(50,45,79,90,88,65,52,79,84,61,52)
d <- pred-po
shapiro.test(d)#ok
```

```
##
##  Shapiro-Wilk normality test
##
## data:  d
## W = 0.95142, p-value = 0.6621
```

```r
t.test(d,alternative = "g")
```

```
##
##  One Sample t-test
##
## data:  d
## t = 3.1128, df = 10, p-value = 0.005504
## alternative hypothesis: true mean is greater than 0
## 95 percent confidence interval:
##  5.77245     Inf
## sample estimates:
## mean of x
##  13.81818
```

Phodnota=0.005<0.05 zamietame H_0 teda prijimame H_1, zlepsili sa. Ina moznost- nerobit rozdiely, ale pridat do prikazu paired=T.

```r
t.test(pred,po,alternative = "g",paired = T)
```

```
##
##  Paired t-test
##
## data:  pred and po
## t = 3.1128, df = 10, p-value = 0.005504
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
##  5.77245     Inf
## sample estimates:
## mean difference
##        13.81818
```